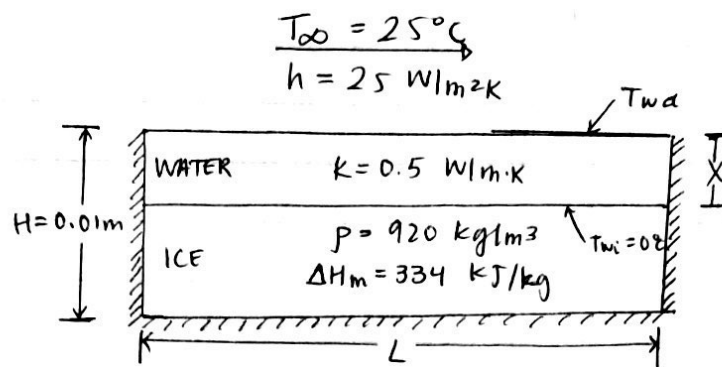


Problem 1:



a) position of water/ice interface as a function of time

E.B. for ice water

$$\dot{E}_{\text{sur}} = \dot{E}_{\text{in}}$$

$$\rightarrow q_{\text{conv}} = \frac{d}{dt} m \Delta H_m$$

$$\rightarrow q_{\text{conv}} = \frac{d}{dt} \rho L^2 x \Delta H_m$$

$$\rightarrow q_{\text{conv}} = \rho L^2 \Delta H_m \frac{dx}{dt}$$

$$* m = \rho V$$

$$m = \rho (L)(L)x$$

$$m = \rho L^2 x$$

E.B. for water $\dot{E}_{\text{out}} = \dot{E}_{\text{in}} \rightarrow q_{\text{cond}} = q_{\text{conv}}$

$$q_b = \frac{k L^2 (T_{w,a} - T_{w,i})}{(x)} = h L^2 (T_{\infty} - T_{w,a})$$

$$\frac{q_b}{L^2} = \frac{(T_{\infty} - T_{w,a})}{1/h}$$

$$\frac{q_b}{L^2} = \rho \Delta H_m \frac{dx}{dt}$$

$$\frac{q_b}{L^2} = \frac{(T_{w,a} - T_{w,i})}{x/k}$$

$$\frac{q_b}{L^2} = \frac{(T_{\infty} - T_{w,i})}{\left(\frac{x}{k}\right) + \left(\frac{1}{h}\right)}$$

$$\int \rho \Delta H_m \left(\frac{x}{k} + \frac{1}{h}\right) dx = \int (T_{\infty} - T_{w,i}) dt$$

$$\rho \Delta H_m \left(\frac{x^2}{2k} + \frac{x}{h}\right) = (T_{\infty} - T_{w,i})(t)$$

$$\rho \Delta H_m \left(\frac{x^2}{2k}\right) + \rho \Delta H_m \left(\frac{x}{h}\right) - (T_{\infty} - T_{w,i})(t) = 0$$

$$x = \frac{-\rho \Delta H_m \left(\frac{1}{h}\right) + \sqrt{\rho^2 \Delta H_m^2 \left(\frac{1}{h^2}\right) - 4 \left(\rho \Delta H_m \left(\frac{1}{2k}\right) - (T_{\infty} - T_{w,i})(t)\right)}}{2 \rho \Delta H_m / 2k}$$

$$x = \frac{-\rho \Delta H_m \left(\frac{1}{h}\right)}{\rho \Delta H_m \left(\frac{1}{k}\right)} + \frac{\sqrt{\rho^2 \Delta H_m^2 \left(\frac{1}{h^2}\right) + 2 \rho \Delta H_m \left(\frac{1}{k}\right) - (T_{\infty} - T_{w,i})(t)}}{\rho \Delta H_m^2 \left(\frac{1}{k^2}\right)}$$

$$x = \left(-\frac{k}{h}\right) + \frac{\frac{k^2}{h^2} + \frac{2k}{\rho \Delta H_m (T_{\infty} - T_{w,i})(t)}}{\rho \Delta H_m^2 \left(\frac{1}{k^2}\right)}$$

b) time required for ice to melt

$$x = h$$

$$(t)_m = \frac{\rho \Delta H_m \left(\frac{H^2}{2k} + \frac{H}{h} \right)}{(T_\infty - T_{w,i})} = \frac{(920)(334)(10)^3 \left(\frac{(0.01)^2}{2(0.5)} + \frac{(0.01)}{25} \right)}{(25 - 0)} = \boxed{6145.6 \text{ seconds}}$$

no resistance:

$$(t)_m = \frac{\rho \Delta H_m \left(\frac{H}{h} \right)}{(T_\infty - T_{w,i})} = \frac{(920)(334)(10)^3 \left(\frac{0.01}{25} \right)}{(25)} = \boxed{4916.5 \text{ seconds}}$$

less time than

Bonus $T_{w,a}$ changes with time. $T_{w,a} = ?$

$$\frac{T_\infty - T_{w,a}}{\left(\frac{1}{k}\right)} = \frac{T_\infty - T_{w,i}}{\left(\frac{x}{k}\right) + \left(\frac{1}{h}\right)} \rightarrow \frac{T_\infty - T_{w,a}}{T_\infty - T_{w,i}} = \frac{\left(\frac{1}{h}\right)}{\left(\frac{x}{k}\right) + \left(\frac{1}{h}\right)} = \frac{1}{\left(\frac{h}{k}\right) + 1}$$

$$= \frac{1}{\left(1 + \left(\frac{h}{k}\right) \left[\left(\frac{x}{h}\right) + \sqrt{\left(\frac{k^2}{h^2}\right) + \left(\frac{2k}{\rho \Delta H_m}\right) (T_\infty - T_{w,i}) (t)} \right] \right)}}$$

$$= \frac{1}{1 + (-1) + \sqrt{\left(\frac{h^2}{k^2}\right) \left(\frac{k^2}{h^2}\right) + \left(\frac{h^2}{k^2}\right) \left(\frac{2k}{\rho \Delta H_m}\right) (T_\infty - T_{w,i}) (t)}}$$

$$= \left(1 + \frac{2h^2 (T_\infty - T_{w,i}) (t)}{k \rho \Delta H_m}\right)^{-1/2}$$

$$x = H$$

$$\frac{T_\infty - T_{w,a}}{T_\infty - T_{w,i}} = \frac{1}{1 + \left(\frac{h}{k}\right) H} \rightarrow \frac{25 - T_{w,a}}{25 - 0} = \frac{1}{1 + \frac{25(0.01)}{(0.5)}}$$

$$\boxed{T_{w,a} = 8.3^\circ\text{C}}$$

2. 100°C , 20°C 65°C

(a) outlet temp. for parallel flow

$$q = \frac{\dot{m}_h c_{ph}}{\dot{m}_c c_p} (T_{h,i} - T_{h,o}) = \frac{\dot{m}_c c_{pc}}{\dot{m}_c c_p} (T_{c,o} - T_{c,i})$$

$$(T_{h,i} - T_{h,o}) = (T_{c,o} - T_{c,i})$$

$$100 - 65 = T_{c,o} - 20 \longrightarrow \boxed{T_{c,o} = 55^\circ\text{C}}$$

(b) counter flow outlet temps $T_{c,o} = ?$, $T_{h,o} = ?$

$$\varepsilon = \frac{q}{q_{\max}} = \frac{35 \dot{m}_c c_p}{80 \dot{m}_c c_p} = 0.4375$$

$$NTU = \frac{-\ln[1 - \varepsilon(1 + Cr)]}{(1 + Cr)} = \frac{-\ln[1 - (0.4375)(1 + 1)]}{(1 + 1)} = 1.04$$

$$\frac{UA}{\dot{m}_c c_p} = \frac{UA}{\dot{m}_h c_p} = NTU = 1.04 \longrightarrow NTU_{\text{parallel}} = NTU_{\text{counter}}$$

$$\varepsilon = \frac{1 - \exp[-NTU(1 - Cr)]}{1 - Cr \exp[-NTU(1 - Cr)]} = \lim_{Cr \rightarrow 1} \frac{\frac{d}{dCr} [1 - \exp[-NTU(1 - Cr)]]}{\frac{d}{dCr} [Cr - \exp[-NTU(1 - Cr)]]}$$

$$\varepsilon = \lim_{Cr \rightarrow 1} \frac{-NTU e^{-NTU(1 - Cr)}}{-NTU e^{-NTU(1 - Cr)} - 1} = \frac{NTU}{NTU + 1} = \frac{1.04}{1.04 + 1} = 0.51$$

$$q = \varepsilon q_{\max} = \varepsilon \dot{m}_c c_p 80 \longrightarrow \frac{q}{\dot{m}_c c_p} = \varepsilon(80) = (0.5097)(80) = 40.78$$

$$40.78 = 100 - T_{h,o}$$

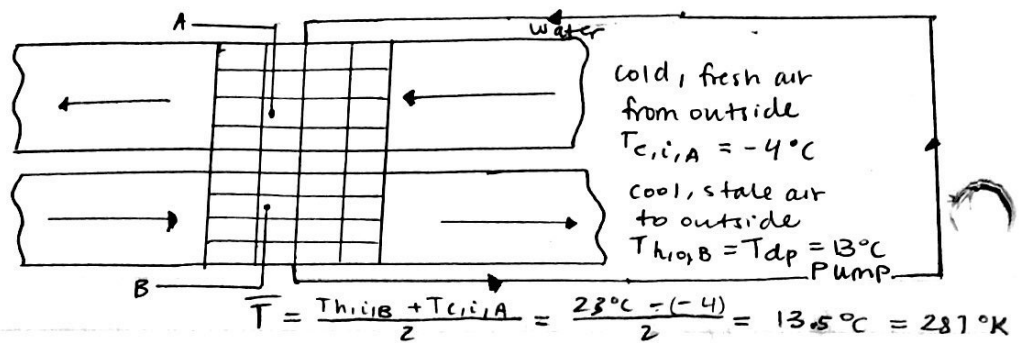
$$40.78 = T_{c,o} - 20$$

$$\boxed{T_{h,o} = 59.22^\circ\text{C}}$$

$$\boxed{T_{c,o} = 60.78^\circ\text{C}}$$

warmed, fresh air to interior

Warm, stale air from interior
 $T_{h,i,B} = 23^\circ\text{C}$



11.29 $\dot{m} = 1.50 \text{ kg/s}$
 $UA = 2500 \text{ W/K}$

Table A.6 for water $6T = 287\text{K}$, $C_p = 4194 \text{ J/kg}\cdot\text{K}$

Table A.4 for air $6T = 283\text{K}$, $C_p = 1007 \text{ J/kg}\cdot\text{K}$

$$C_{\max} = C_{\text{air}} = \dot{m}_{\text{air}} C_{p,\text{air}} = (1.5 \text{ kg/s})(1007 \text{ J/kg}\cdot\text{K}) = 1510 \text{ W/K}$$

$$q = q_A = q_B = C_{\max} (T_{h,i,B} - T_{h,o,B}) = (1510 \text{ W/K})(23^\circ\text{C} - 13^\circ\text{C}) = \boxed{15,100 \text{ W}}$$

$$NTU = \frac{UA}{C_{\min}} = \frac{(2500 \text{ W/K})}{C_{\min}}$$

$$q = \epsilon C_{\min} (T_{h,i,A} - T_{c,i,A}) \rightarrow (15,100 \text{ W}) = \epsilon C_{\min} (T_{h,i,A} - 4^\circ\text{C})$$

$$q = \epsilon C_{\min} (T_{h,i,B} - T_{c,o,B}) \rightarrow (15,100 \text{ W}) = \epsilon C_{\min} (23^\circ\text{C} - T_{c,o,B})$$

$$q = C_{\min} (T_{c,o,B} - T_{c,i,B}) \rightarrow (15,100 \text{ W}) = C_{\min} (T_{c,o,B} - T_{c,i,B})$$

$$C_r = \frac{C_{\min}}{C_{\max}}$$

$$T_{c,o,B} = T_{h,i,A}$$

$$T_{h,o,A} = T_{c,i,B}$$

$$\epsilon = 1 - \exp\left[-\frac{UA}{C_{\max}}\right] = 1 - \exp\left[-\frac{(2500 \text{ W/K})}{(1510 \text{ W/K})}\right] = 0.809$$

with trial + error

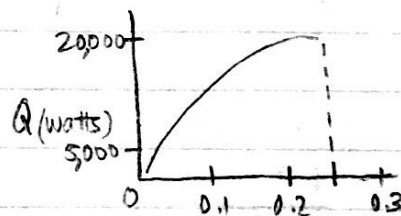
$$\dot{m}_{\text{water}} = \frac{C_{\min}}{C_{p,\text{water}}} = \frac{776}{4194 \text{ J/kg}\cdot\text{K}} = 0.185 \text{ kg/s}$$

$$\dot{m}_{\text{water,max}} = \frac{C_{\max}}{C_{p,\text{water}}} = \frac{1510}{4194} = 0.36 \text{ kg/s}$$

$$Q = \epsilon C_{\text{air}} (23 - 9.5)$$

$$Q = (0.809)(1510)(23 - 9.5)$$

$$Q = 16,496.9 \text{ W}$$



the max heat transfer rate is at
 $\dot{m}_{\text{water}} \approx 0.25 \text{ kg/s}$

11.1 $U = 300 \text{ W/m}^2\cdot\text{K}$

$$R''_{f,i} = a_i t, \quad R''_{f,o} = a_o t$$

$$a_i = 2.5 \times 10^{-11} \text{ m}^2\cdot\text{K/W}\cdot\text{s}$$

$$a_o = 1.0 \times 10^{-11} \text{ m}^2\cdot\text{K/W}\cdot\text{s}$$

25%

$$\frac{1}{U_f} = \frac{1}{U_c} + R''_{f,i} + R''_{f,o}$$

$$= \left(\frac{1}{U_c}\right) + (a_{o,i} + a_{o,o})(t)$$

$$t = \frac{\left(\frac{1}{U_f}\right) - \left(\frac{1}{U_c}\right)}{(a_{o,i} + a_{o,o})} = \frac{\left(\frac{1}{0.75(300 \text{ W/m}^2\cdot\text{K})}\right) - \left(\frac{1}{(300 \text{ W/m}^2\cdot\text{K})}\right)}{(2.5 \times 10^{-11} \text{ m}^2\cdot\text{K/W}\cdot\text{s}) + (1.0 \times 10^{-11} \text{ m}^2\cdot\text{K/W}\cdot\text{s})}$$

$$(31.75(10)^6 \text{ seconds}) \left(\frac{1 \text{ hour}}{3600 \text{ seconds}}\right) \left(\frac{1 \text{ day}}{24 \text{ hours}}\right) \left(\frac{1 \text{ year}}{365 \text{ days}}\right)$$

$$t = 1.01 \text{ years}$$